

E-Commerce Dataset: SQL Analysis & Insights

A consultant-style deep dive into pricing efficiency, discount patterns, and category-level performance across a PostgreSQL e-commerce database

1

Data Source

PostgreSQL E-Commerce Dataset — 5 CSV category tables unified

2

Scope

15 category-price tier combinations across 5 major product segments

3

Objective

Data-backed merchandising decisions through pricing & discount intelligence

Baby & Maternity

4,412 products

Beauty & Health

1,889 products

Men's Clothing

1,889 products

Shoes

4,360 products

Women's Clothing

3,889 products

Executive Summary: Dataset Overview

Scope & Structure

Database: PostgreSQL — E-Commerce Dataset with multiple CSV-sourced category tables covering Baby & Maternity, Beauty & Health, Men's Clothing, Shoes, and Women's Clothing.

Core view analyzed: `clean_ecommerce_products` — a unified view consolidating all category tables into a single analytical layer.

Total combinations: 15 category-price tier combinations across 5 major product categories, enabling cross-sectional comparison.

Key Analytical Dimensions

main_category

Product category label
from source CSV tables

price_tier

Budget / Mid-range /
Premium segmentation

avg_discount_per cent

Mean discount rate per
category-tier group

avg_original_pric e & avg_selling_price

Price spread capturing
markdown depth

product_count

Volume proxy for
category market weight

 **Analysis goal:** Identify pricing efficiency, discount patterns, and category-level performance to drive data-backed merchandising decisions.

Category-Wise Pricing Summary



Category Performance at a Glance

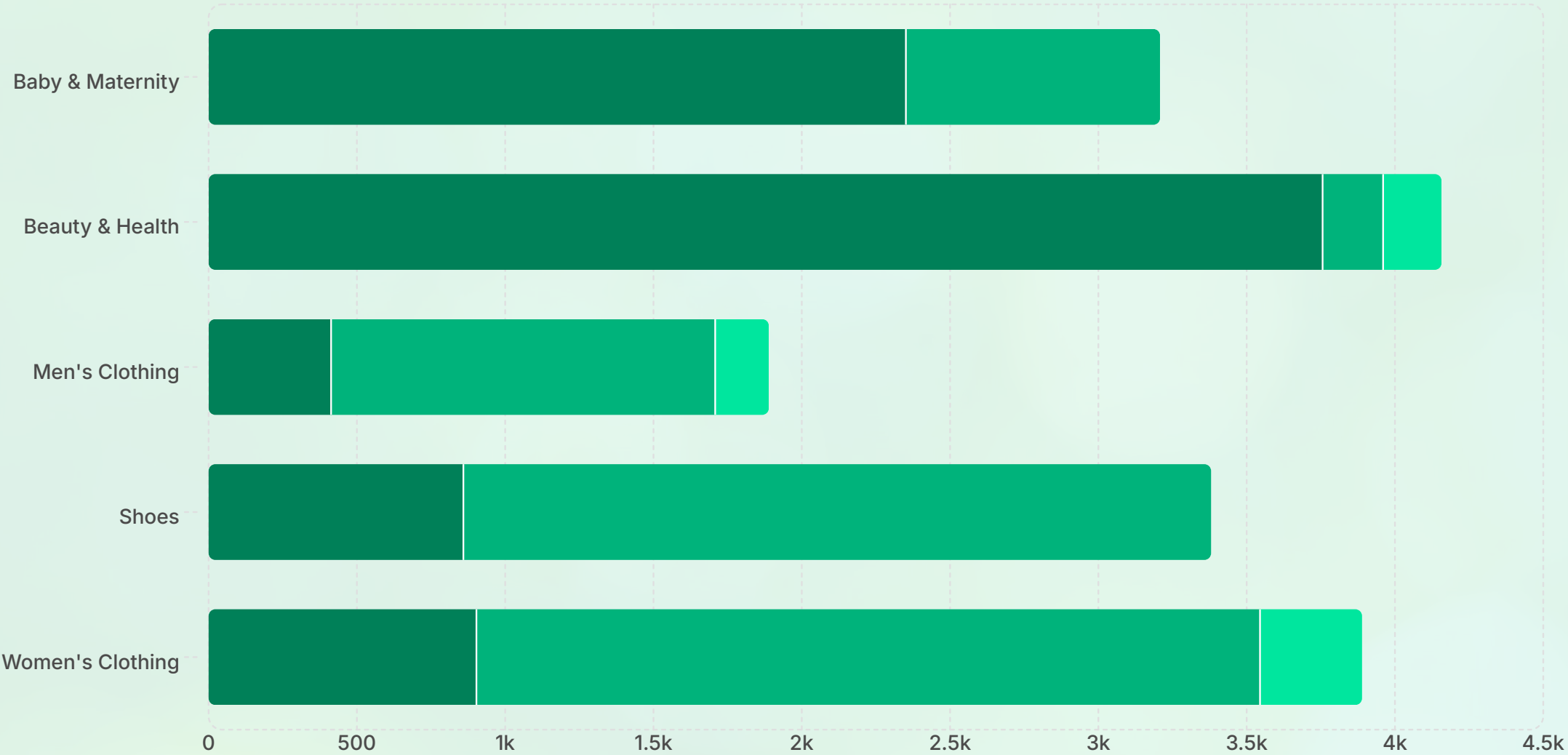
Category	Products	Avg Discount
Shoes	4,360	26.92%
Men's Clothing	1,889	17.53%
Baby & Maternity	4,412	10.66%
Women's Clothing	3,889	12.00%
Beauty & Health	1,889	12.00%

Shoes leads in both volume and discount aggression at **26.92%** avg markdown. **Women's Clothing** posts the lowest avg selling price (\$5.37) despite high volume — a margin efficiency concern. **Beauty & Health** sustains a premium positioning with only a 18.6% price gap between original and selling price.

Price Tier Breakdown: Budget vs Mid-Range vs Premium

■ Budget ■ Mid-range ■ Premium

Category



Budget Tier

Beauty & Health dominates budget volume with **3,756 products** at avg \$15.03, followed by Baby & Maternity (**2,351** at \$14.75). This indicates strong price-sensitive consumer demand in both segments.

Mid-Range Tier

Women's Clothing (**2,642**) and Shoes (**2,521**) anchor the mid-range. Shoes mid-range avg price reaches \$24.91 — the highest of any tier-category combination. **Men's Clothing mid-range avg shows \$0.00** — a data quality flag requiring investigation.

Premium Tier

Only three categories have premium presence. Beauty & Health leads at avg \$24.56 for **197 products**, while Women's Clothing Premium paradoxically shows avg \$2.43 — likely a classification anomaly. No premium tier exists for Baby & Maternity or Shoes.

📌 **Key insight:** Mid-range tier dominates product count across all categories — indicating the platform skews toward value-for-money positioning rather than aspirational or luxury merchandising.

Discount Bracket Segmentation

Discount brackets defined via SQL CASE logic: **Low Discount** (<20%) | **Moderate Discount** (20–49%) | **Heavy Discount** (≥50%)



Baby & Maternity



Low: 624 | Moderate: 182 | **Heavy: 419** — significant heavy-discount tail despite the lowest avg discount rate of any category. Signals inconsistent pricing policy.

Beauty & Health



Low: 1,689 | Moderate: 415 | **Heavy: 0** — zero heavy discounting. Premium brand protection strategy is intact and working.

Men's Clothing



Low: 3,223 | Moderate: 0 | **Heavy: 0** — entirely low-discount. Stable, predictable pricing with no markdown exposure.

Shoes



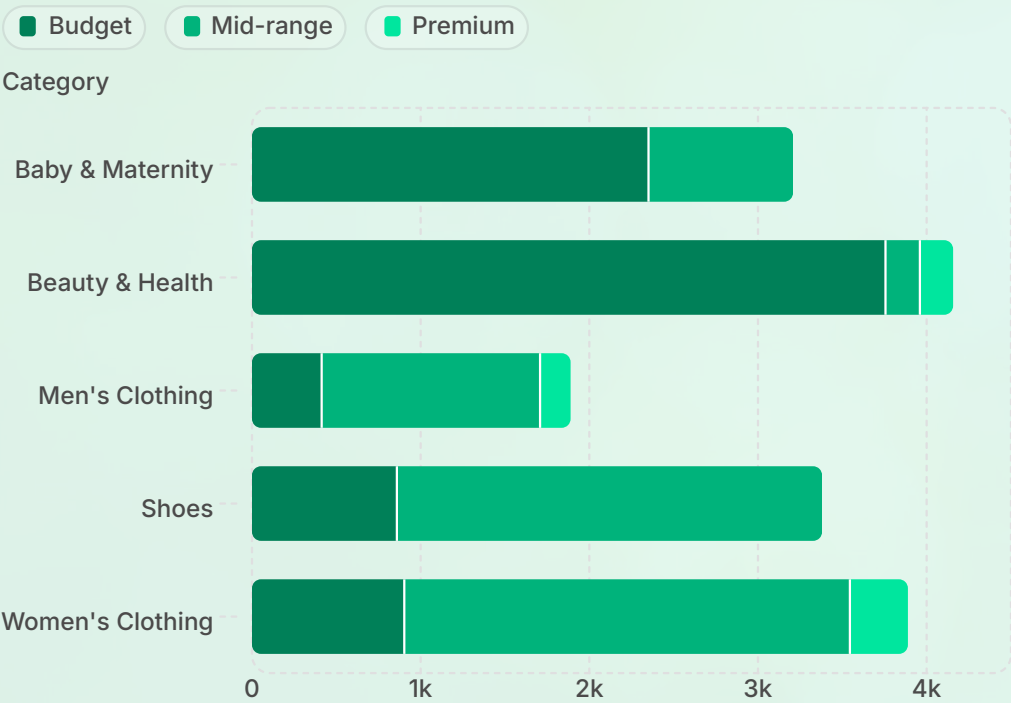
Low: 685 | Moderate: 0 | **Heavy: 0** — concentrated in low bracket despite having the highest avg discount %. Bracket vs. average divergence worth investigating.

Women's Clothing



Low: 4,181 | Moderate: 367 | **Heavy: 71** — dominant low-discount pool with minor but present markdown activity at the tail.

Price Tier Breakdown: Budget vs Mid-Range vs Premium



Tier-Level Observations

Baby & Maternity

Budget: 2,351 @ \$14.75 avg ·
Mid-range: 857 @ \$15.00 ·
No Premium SKUs — entirely
sub-premium range

Beauty & Health

Budget: 3,756 @ \$15.03 ·
Mid-range: 204 @ \$4.65 ·
Premium: 197 @ \$24.56 —
bi-modal price distribution

Men's Clothing

Mid-range: 1,295 · Budget:
413 · Premium: 181 — all tiers
show \$0.00 avg price (data
quality flag)

Shoes

Mid-range: 2,521 @ \$24.91 ·
Budget: 859 @ \$20.16 · No
Premium tier — mid-range
clearly dominant

Women's Clothing

Mid-range: 2,642 @ \$6.65 ·
Budget: 903 @ \$10.15 ·
Premium: 344 @ \$2.43 —
inverted pricing anomaly in
premium

📌 **Platform-level insight:** Mid-range tier dominates product count across all categories — confirming the platform skews toward value-for-money positioning. Premium penetration remains thin, representing an underexploited revenue lever.

Key Findings & Strategic Recommendations

Shoes — Review Markdown Strategy

Highest volume (4,360 products) and highest avg discount (26.92%) is a margin erosion risk. Audit markdown triggers and introduce dynamic pricing guardrails to protect gross margin without sacrificing conversion.

Baby & Maternity — Standardize Discounting

419 products in the heavy-discount bracket despite a low 10.66% avg discount rate signals inconsistent pricing execution. Implement a unified discount policy and flag outlier SKUs for repricing review.

Beauty & Health — Hold the Line

Zero heavy discounts, stable avg selling price of \$10.00, and a clear premium presence (197 products at \$24.56). Maintain brand integrity — resist promotional pressure and protect the premium tier positioning.

Men's Clothing — Introduce Premium Tier

Entirely low-discount and mid-range dominant with 1,295 products — stable but under-monetized. Opportunity to introduce a curated premium assortment to lift Average Selling Price (ASP) and attract higher-LTV customers.

Women's Clothing — Drive Basket Size

Largest low-discount pool (4,181 products) but lowest avg selling price (\$5.37). High volume at low price requires bundling, upsell, and cross-sell strategies to improve revenue per transaction and basket size.

Next Steps

1

Time-Series Discount Trends

Build temporal analysis to track markdown velocity over time per category

2

Customer Ratings Integration

Join ratings data to price tiers to assess quality-price perception alignment

3

Cohort Analysis by Price Tier

Identify highest-LTV customer segments to prioritize inventory and promotions